

Thermodynamics

Introduction to thermodynamics - Energy transfer heat, work and mass.

Thermodynamics:-

Deals with the transfer of energy from one form to another or from one place to another.

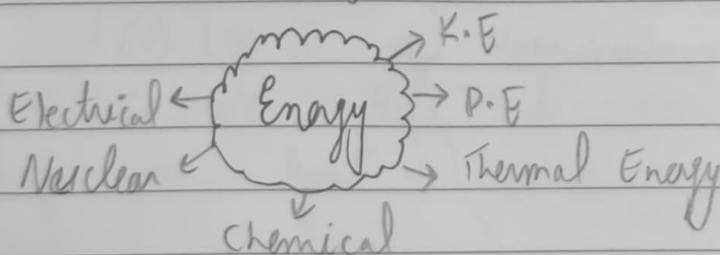
Energy:-

A quantitative property that must be transfer from a body or physical system to perform work or transfer heat.

Heat:

Form of energy that is transferred between two bodies due to difference in the temperature.

→ Rate of movement of molecules



Energy

Macroscopic Energy



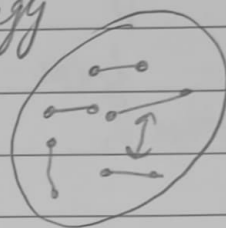
K.E, P.E

Microscopic Energy



Internal Energy

{
Translational Energy
Vibrational Energy
Rotational Energy
Binding force



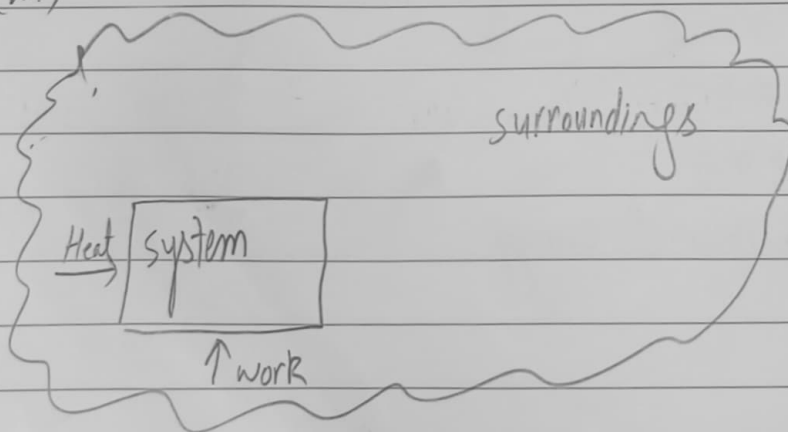
molecules /
Atom

$$\text{Total Energy} = K.E + P.E + U$$

$$= \frac{1}{2}mv^2 + mgh + U$$

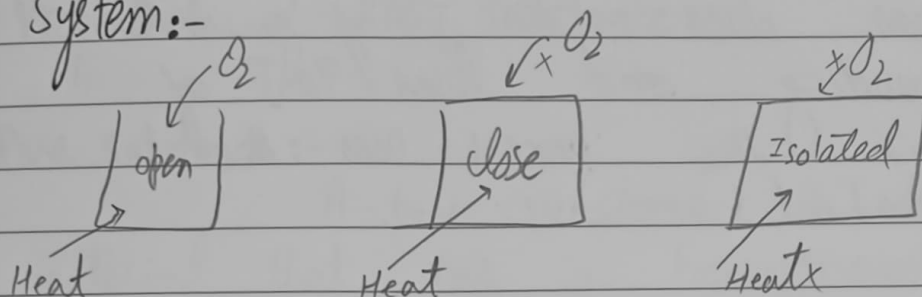
Mechanism of Energy transfer

- 1- Work Done (W) → Joule → Mechanical energy
- 2- Heat (q) → Joule → Thermal Energy
- 3- Mass (m)
flow





System:-



~~$\Delta U = q + w$ (Chemistry)~~

$\Delta U = q - w$ (Physics) $\rightarrow$ use this in our course

Chemistry:-

$W = -ve \rightarrow$ Work done by the system

$W = +ve \rightarrow$ Work done on the system

Physics:-

$W \rightarrow -ve \rightarrow$ Work done ON the system

$W \rightarrow +ve \rightarrow$ Work done BY the system

Heat:-

$q \rightarrow +ve$ (absorb) $\rightarrow$ endothermic

$q \rightarrow -ve$ (release) $\rightarrow$ exothermic

Example:-

① 100 J of work is done on the system
 U increases by 74 J. How much energy is transferred as heat?

$\Delta U = q - w$

$-74 = q - 100$

$q = -26 \text{ J}$

② A sample of gas does 150 J of work against its surrounding and loses 90 J of internal energy in the process. Does the gas lose or gain heat?

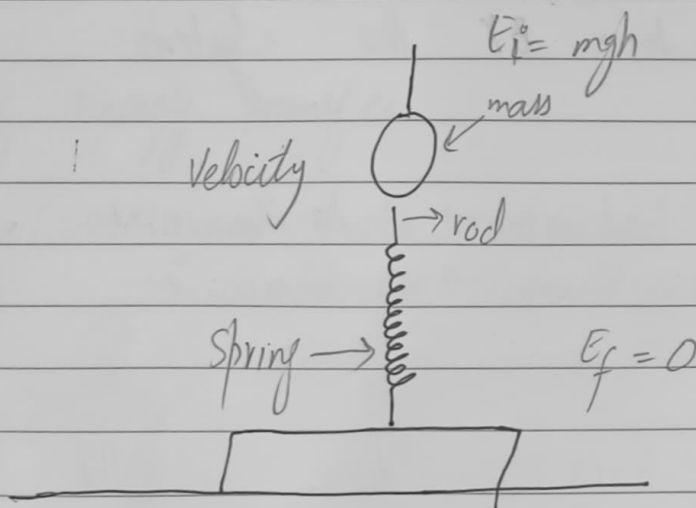
$$\Delta U = q - w$$

$$q = \Delta U + w$$

$$q = -90 + 150$$

$$q = +60 \text{ J}$$

③



Properties of Pure substance

Pure substance :-

A pure substance is defined as a material that has a homogeneous and invariable chemical composition.

Homogeneity

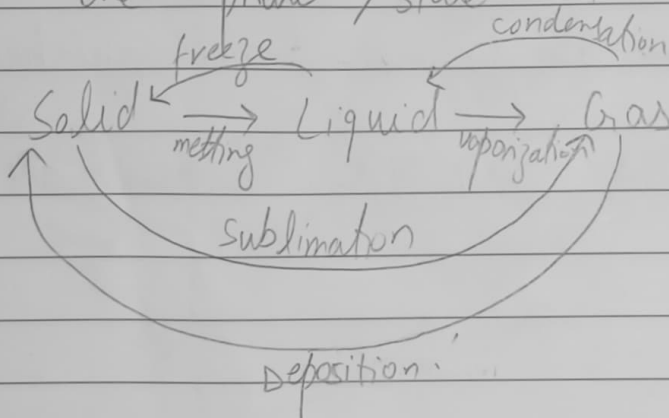
The substance is the same throughout, with no regions having a different composition.

Invariable composition.

The chemical composition doesn't change even when the substance transitions between different phases.

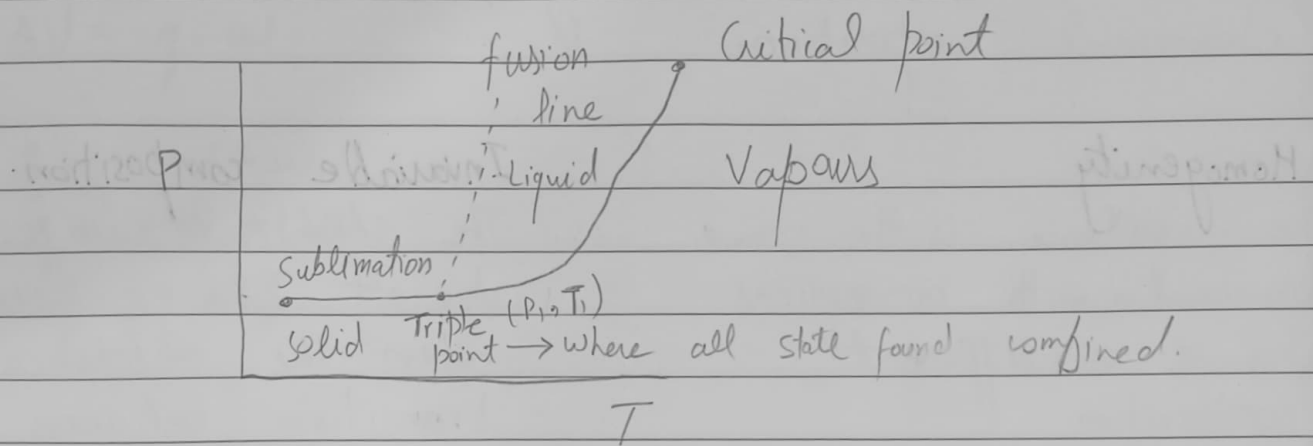
Phase Change and Diagram

Phase change of material occurs when it transitions from one phase/state to another.



Saturation Temperature :-

The specific temperature at which a substance undergoes a phase change under a given pressure. (vaporization, condensation)



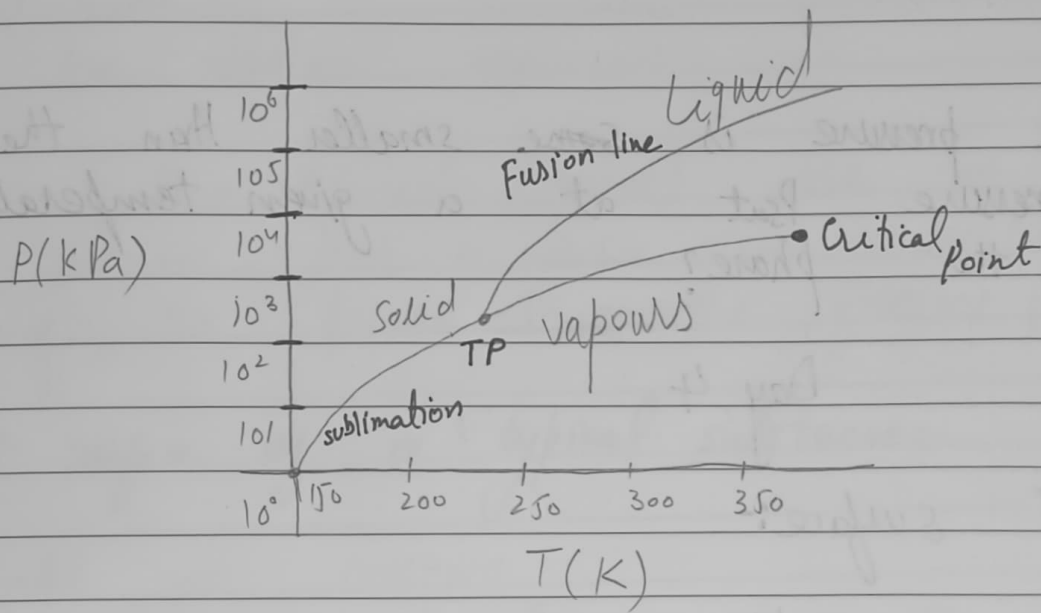
Triple point of water occurs at 0.01°C , 0.6117 kPa

Critical Point marks the end of vaporization curve on a phase diagram. Beyond this point liquid & gaseous phase becomes indistinguishable.

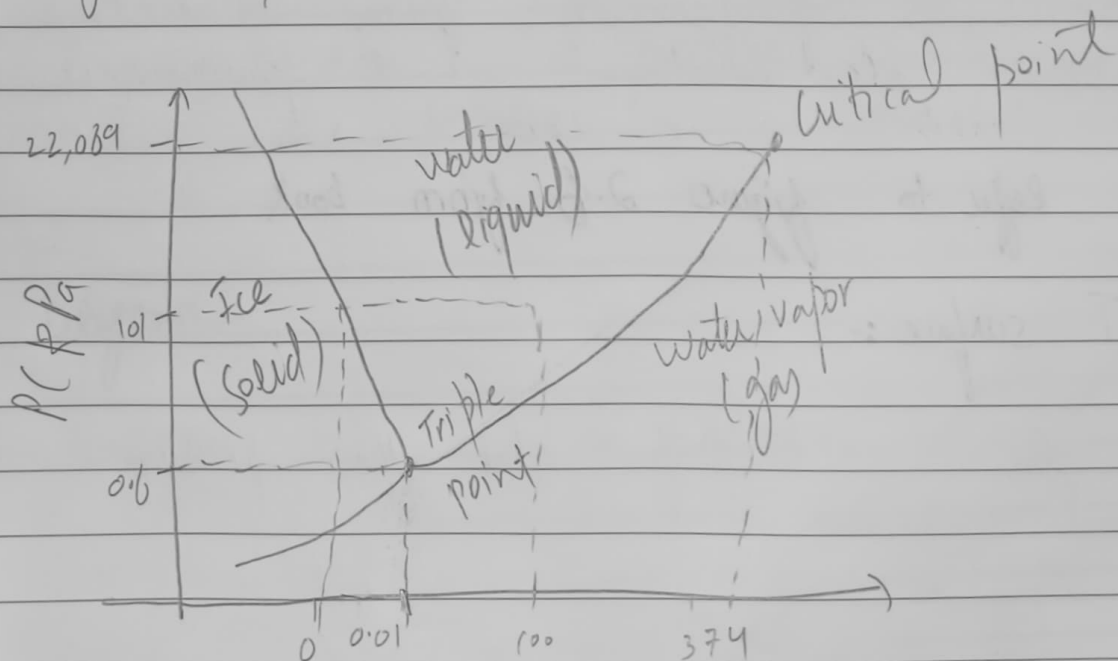
Critical Point of water 374.14°C and 22.09 MPa



Phase diagram of CO_2



Phase diagram of H_2O





Question: what is the lowest temperature at which water can remain in the liquid form?

0.01°C

Question: If the pressure is ~~same~~ smaller than the smallest pressure P_{sat} at a given temperature T , what is the phase?

Day 4

The P - V - T surface:-

- critical for understanding how substances behave under different conditions. (P, V, T)
- Applications → Refrigeration, power generation, etc.
compressor etc.

Refer to figure 2.6 from book

P - V - T surface:-

- Captures the relationship b/w the pressure, volume and temperature

P-v-T surface of water:-

- Ice → has larger volume than liquid state.
- liquid water
- (water has abnormal behaviour)

The fusion line (Solid-liquid) line on pressure-Temp diagram has a negative slope indicating the increasing the pressure lowers the freezing point.

P-v-T surface of a typical substance:-

- shrinks on freezing
- Solid phase have lower volume than liquid phase.

The fusion line (Solid-liquid) line on pressure-Temp diagram has a positive slope indicating the increasing the pressure the freezing point.

Phase Diagram:-

- P-T diagram
- Fusion line
- Sublimation line
- Triple Point
- Critical Point



T-V diagram:-

- Show how volume changes with temperature at constant pressure illustrating phase change from liquid to vapour.
- Depicts the boiling phase.

P-V diagram:-

- Shows how pressure changes with specific volume at constant temperature.

Independent properties:-

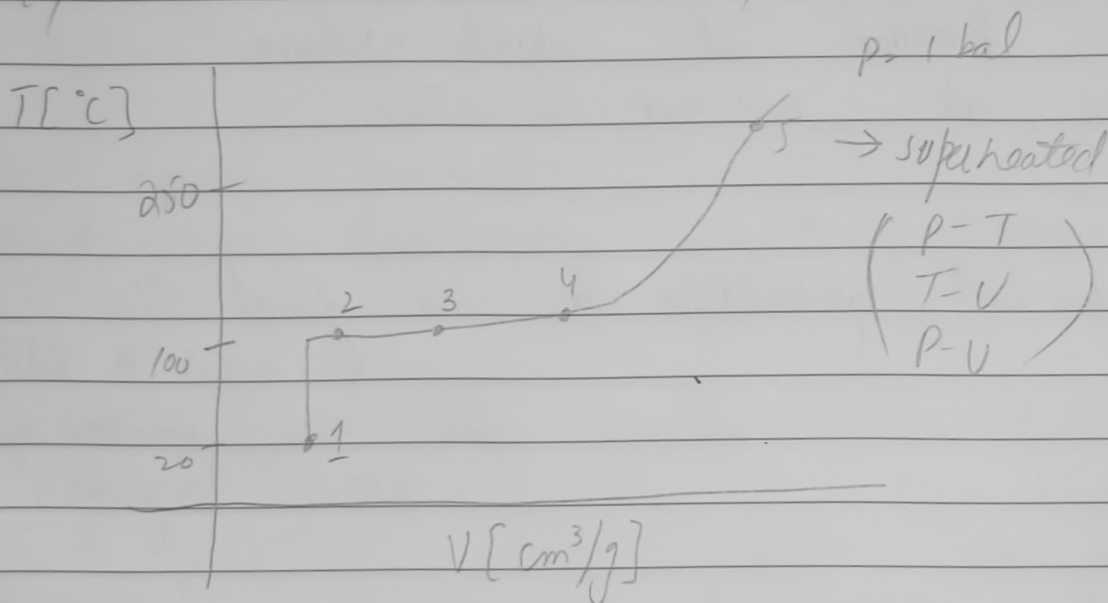
The states of substance can be defined by two independent properties. However, in the two phase region (during boiling or condensation), pressure and temperature are independent because they are fixed at a saturated point for a given pressure-temperature pair.

Tables of Thermodynamics properties

State 1	State 2	State 3
(liquid)	(liquid)	(liquid/vapour)
$P = 1 \text{ bar}$	$P = 1 \text{ bar}$	$P = 1 \text{ bar}$
$T = 20^\circ\text{C}$	$T = 100^\circ\text{C}$	$T = 100^\circ\text{C}$
		→ vapour
		→ liquid

Heat

State 4 (vapour)	State 5 (vapour)
$P = 1 \text{ bar}$	$P = 1 \text{ bar}$
$T = 100^\circ\text{C}$	$T = 250^\circ\text{C}$



Purpose:-

Thermodynamics tables provide essential data for various substance that describe their thermodynamic properties (like temperature, pressure, volume V , specific volume, internal energy like enthalpy and entropy)

Saturation temperature:-

Temperature at which phase change occurs at constant Pressure

Saturation Pressure:-

Pressure at which phase change occurs at constant Temperature.

⇒ with 2 independent properties we can find the rest

- Temperature (T)
- Pressure (P)
- Specific volume (v)
- Specific Internal energy (u)
- Specific Enthalpy (h)
- Specific entropy (s)

- Linear Interpolation
- Polynomial Regression
- Moving average
- Lagrange Interpolation.

Saturated liquid and Vapour tables

Table B1.1 :-

v_f = Specific volume of the saturation liquid
 v_g = Specific volume of the saturation vapours
 v_{fg} :- Difference between v_g & v_f

$$v_{fg} = v_g - v_f$$

First Column:- Temperature

Second Column:- Corresponding Saturation Pressure (P_{sat})

Table B 1.2 :-

→ contains the same information as B 1.1 but organized pressure

→ Useful



Example 2.1:-

a) The saturation pressure is 198.5 kPa , so we have a compressed liquid, that is above the saturation line for 120°C . We could also have entered with 500 kPa and found the saturation temperature as 151.86°C , so we would say that it's a subcooled liquid.

b) $v_f = 0.00106 < v < v_g = 0.89186 \text{ m}^3/\text{kg}$
so the state is a two-phase mixture of liquid and vapour. The state is to the left of the vapour state and to the right of the saturated liquid state.

Question # 25:-

a) $T = 260^\circ\text{C}$, $P = 5 \text{ MPa}$

$$P_{\text{sat}} = 4.688 \text{ MPa}$$

$$P_{\text{sat}} (4.688 \text{ MPa}) < P (5 \text{ MPa})$$

the phase is compressed liquid.

b)

$$T = -2^\circ\text{C}, P = 100 \text{ kPa}$$

$$T_{\text{freezing point}} = 0^\circ\text{C}$$

$T_{\text{freezing point}} (0^\circ\text{C}) > T (-2^\circ\text{C})$, the phase is solid (ice)



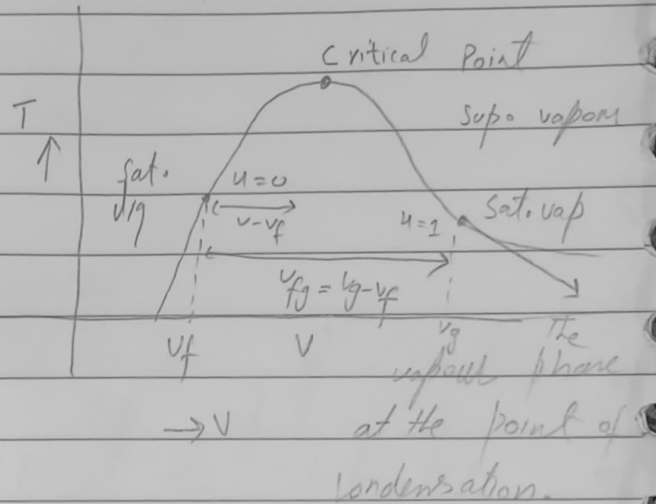
Day 6

The two phase state:-

A mixture of two phases of the same substance, such as liquid and vapour, in equilibrium at the same temperature (T) and Pressure (P)

Specific volume:-

Volume occupied by a unit mass of a substance. Measure of how much space one "kg" of substance occupies



$$v = \frac{V}{m}$$

$$V = mv$$

∴ if total volume is given divide by the mass then we obtain specific volume.

Specific Volume in Two-Phase Mixture:-

Total volume of two-phase mixture is the sum of the volume of the liquid (V_{liq}) and the vapour (V_{vap})

$$V_{total} = V_{liq} + V_{vap} = m_{liq} v_f + m_{vap} v_g$$

Avg. specific volume:-

$$v = \frac{V}{m} = \frac{m_{\text{liq}} v_f}{m} + \frac{m_{\text{vap}} v_g}{m} \quad \because \frac{m_{\text{liq}}}{m} = (1-x) \text{ and } \frac{m_{\text{vap}}}{m} = x$$

$$V = (1-x) v_f + x v_g$$

$$v = (1-x) v_f + x v_g = v_f - x v_f + x v_g \Rightarrow v = v_f + x (v_g - v_f)$$

$$v = v_f + x v_{fg}$$

Quality:

It is the ratio of mass of vapour to the total mass in a two phase mixture

Range: $0 \leq x \leq 1$

$x=0$:- sat. liquid

$x=1$:- sat. vapour



Mon Tue Wed Thu Fri Sat Sun

Date: / /

Problem:-

Calculate the specific volume of a saturated mixture of water at 200°C with a quality of 70%.

Refer: Table B 1.1

Solution:-

$$v = x v_{fg} + v_f = x (v_g - v_f) + v_f$$

$$= (0.7 (0.12736 - 0.001156) + 0.001156)$$

$$= 0.0894988 \text{ m}^3/\text{kg}$$

↳ upto 3 decimal point.

Compressed Liquid:-

A liquid state at a pressure higher than the saturation pressure for a given temperature OR at a temperature lower than the saturation temperature for a given pressure.

$P > P_{\text{sat}}$ at a given T .

$T < T_{\text{sat}}$ at a given P .

$v < v_f$ at a given T & P .

Internal energy
 $v < v_f \rightarrow$

Enthalpy.
 $h < h_f \rightarrow$

Super heated vapour:

A liquid at a Pressure lower than the Saturation pressure for a given temperature OR at a temperature higher than the saturation temp for a given pressure

$P < P_{sat}$ at a given T
 $T > T_{sat}$ at a given P
 $v > v_g$ at a given T or P

$v > v_g \rightarrow$

$h > h_g$

Problem 2.25

Determine the phase of water at

- a) $T = 260^\circ\text{C}$, $P = 5 \text{ MPa}$
- b) $T = -2^\circ\text{C}$, $P = 100 \text{ kPa}$



Table

a) Refer B.1.1

$$P = 4688.6 \text{ kPa}$$

$$= 4.688 \text{ MPa}$$

$$T_{\text{sat}} = 263.19^\circ\text{C}$$

$$T < T_{\text{sat}}$$

$$260^\circ\text{C} < 263.19^\circ\text{C}$$

$$P_{\text{sat}} < P$$

compressed liquid

b) Table B.1.5

$$T = -2^\circ\text{C}$$

$$P = 100 \text{ kPa}$$

$$P = 0.5177 \text{ kPa}$$

$$P_{\text{sat}} < P$$

Day 7

Date

Range : $0 \leq x \leq 1$

Problem :-

calculate the specific volume of a saturated mixture of water at 200°C with a quality of 70%.

Refer : Table B1.1

Sol :-

$$\begin{aligned} V &= x v_{fg} + v_f \\ &= x(v_g - v_f) + v_f \\ &= 0.7(0.12736 - 0.001156) + 0.001156 \\ &= 0.0894988 \text{ m}^3/\text{kg} \end{aligned}$$

DAY-7

Ideal Gas Law:-

Describes the behaviour of gases under certain conditions. It provides simple correlation among pressure, temperature (T) and specific volume (V). It is an approximation and works best under high temperature, low density and low pressure.

Ideal Gas Equation of state :-

$$PV = RT$$

For calculations involving mass / moles :-

$$PV = mRT = nRT$$

$$R = \frac{\bar{R}}{M}$$

Date

$$\bar{R} = 8.3145 \frac{\text{kJ}}{\text{K mol K}}$$

$$PV = mRT = n\bar{R}T$$

$$n = \frac{m}{M}$$

n is the number of moles

- Low pressure
- High temperature
- Low density

Mass flow rate — Ideal Gas Law

QUESTION: (Table 8.5)

Mass of air?

Temp: 25°C , $P = 100 \text{ kPa}$, Molecular mass = 18.015 kg/mol

$$PV = mRT \quad (R = 0.2875 \text{ kJ/kg}\cdot\text{K})$$

$$m = \frac{PV}{\bar{R}T}$$

$$m = \frac{100 \times 240}{0.287 \times 298}$$

$$R = \frac{\bar{R}}{M} = \frac{8.3145}{28.97} = 0.287$$

$$\bar{R} = 8.3145$$

QUESTION:

Tank has a volume = 0.5 m^3

Date

$m = 10 \text{ kg}$ for ideal gas.
molecular mass = 24, $T = 25^\circ \text{C}$

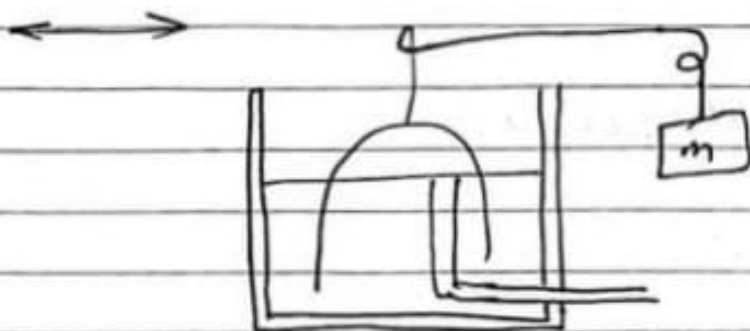
Sol

$$PV = mRT$$

$$P = \frac{mRT}{V}$$

$$= \frac{10 \times 0.3456 \times 298}{0.5}$$

$$= 2064 \text{ Pa}$$



A gas bell is submerged with its mass counterbalanced with rope and pulleys. The pressure is 105 kPa and the T is 21°C . Volume increase is measured to be 0.75 m^3 over a period of 185 seconds. What is the volume flow rate and mass flow rate.

$$R = 0.1889 \text{ kJ/kg.K}$$

$$PV = mRT$$

$$\dot{V} = \frac{0.75 \text{ m}^3}{185 \text{ s}} = \frac{\Delta V}{\Delta T}$$

$$= 0.00405 \text{ m}^3/\text{s}$$

Date

$$P\dot{V} = \dot{m} RT$$

$$\dot{m} = \frac{P\dot{V}}{RT}$$

$$= \frac{(105 \text{ kPa})(0.004 \text{ m}^3/\text{s})}{(0.1889 \text{ kg/mol})(294 \text{ K})}$$

$$= 7.6 \text{ mmol/s}$$

$$= 0.00766 \text{ kg/s}$$

DAY-8

Compressibility Factor (z):

To account for the deviations in ideal gas law, compressibility factor is used

$$z = \frac{PV}{RT}$$

$$PV = zRT$$

($z = 1 \rightarrow$ Gas is ideal)

($z \neq 1 \rightarrow$ Non-ideal behaviour)

if

$z < 1 \rightarrow$ attractive forces b/w molecules dominate causing the gas to be more compressible.

$z > 1 \rightarrow$ Repulsive forces dominate $\rightarrow$ gases resist compression

$\rightarrow$ At low pressures, z approaches 1 as pressure approaches 0

$\rightarrow$ At high temperatures, for many gases like nitrogen and air, z remains close to 1 upto pressures of around 10 MPa.

This means ideal gas law is still a good approximation even at moderate pressures.

Day 8

Compressibility Factor (Z)

To account for the deviations in ideal gas, compressibility factor is used

$$Z = \frac{PV}{RT}$$

$$PV = ZRT$$

($Z=1 \rightarrow$ Gas is ideal)

$Z \neq 1 \rightarrow$ Non ideal behaviour
(Gas is real)

If $Z < 1 \rightarrow$ attractive forces b/w molecules dominate causing the gas to be more compressible

$Z > 1 \rightarrow$ Repulsive force dominate $\rightarrow$ gases resist compression.

$\rightarrow$ At low pressure, Z approaches 1 as pressure approach zero

$\rightarrow$ At high temperature, for many gases like nitrogen and air, Z remains close to 1 upto pressure of around 10 MPa.

This means ideal gas law is still a good approximation even at moderate pressure.

Reduce pressure:-

$$P_r = \frac{P}{P_c}$$

P_c : critical Pressure

Reduce Temperature:

$$T_r = \frac{T}{T_c}$$

T_c = Critical Temperature

These reduced properties allows us to plot z as function of P_r, T_r

If $P < P_c \rightarrow$ Gas behave nearly ideally.

$P \rightarrow P_c \rightarrow$ Gas behave non ideally.

T

P_{sat}

V

190

u_1

V_1

200

u_2

V_2

Day 9

Chapter # 3

First law of thermodynamics for control mass.

→ Control Mass: A fixed mass system mass doesn't change.

→ Control Volume: A spatial region where mass can enter and exit.

→ Heat: Energy transfer due to temperature difference

→ Work transfer: Energy transfer due to the force applied over a distance.

Energy equation for control mass:-

Instantaneous Rate form

$$E_{cv} = \dot{Q} - \dot{W}$$

$$\frac{dE_{cv}}{dt} = \dot{Q} - \dot{W}$$

→ $\dot{E}$: Rate of change of total energy within the control mass.

Sign Convention:

+ve:

Heat ($\dot{Q}$) → when heat is entering the system

Work ($\dot{W}$) → work done by the system

Energy Equation for finite changes:-

Consider total energy changes over a process from state 1 $\xrightarrow{\text{to}}$ state 2

• Starting point

$$dE_{cv} = \delta Q - \delta W$$

Integration over time

$$\int_{t_1}^{t_2} dE_{cv} = \int_{t_1}^{t_2} \delta Q - \int_{t_1}^{t_2} \delta W$$

$$E_2 - E_1 = Q_{1-2} - W_{1-2}$$

↳
total heat added

from state 1 $\xrightarrow{\text{to}}$ state 2

Energy Equation (Detailed)

$$E_{\text{Total}} = U + K.E + P.E$$

$$E_2 - E_1 = (U_2 - U_1) + \frac{1}{2} m (v_2^2 - v_1^2) + mg(h_2 - h_1)$$

Example:

Heat absorb = +500 J

W = +200 J

Find U = ?

$$\begin{aligned} \Delta U &= q - w \\ &= 500 - 200 \\ &= 300 \text{ J} \end{aligned}$$

State function

→ Depend only on the state of the system e.g. U, E

Path function

Depend on the specific process path taken
eg. $Q_{(1-2)}$, W

Pressure Volume Work:-

$$\delta W = PdV = \frac{F}{A} dA \cdot ds$$

$$\delta W = F \cdot ds$$

Iso thermal: → T → same

$$PV = nRT$$



Isothermal equation of an ideal gas in term of work:

→ Isothermal Process

Constant temperature

$$PV = nRT$$

→ Isobaric Process

Constant Pressure

$$W = P(V_2 - V_1)$$

→ Adiabatic Process

Isolated material → $Q = 0$

Day 10

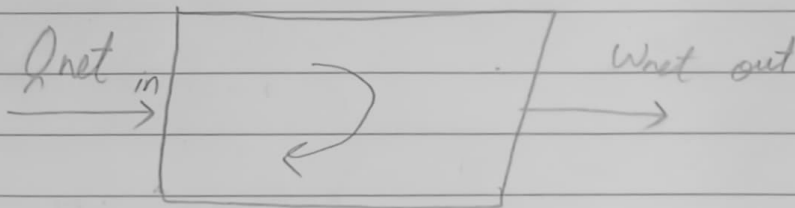
First law of thermodynamics - Control Mass and cyclic Process.

$$0 = \oint \delta Q - \oint \delta W$$

It implies that there is no net change in the total energy of control mass and the rate of change is zero

$$\oint \delta Q = \oint \delta W \rightarrow \text{cyclic integral of the work done.}$$

cyclic integral of heat transfer





Example Problem # 1

250 kJ of heat is absorb and ^{release} 150 kJ of heat

$$Q_1 = 250 \text{ kJ}$$

$$Q_2 = -150 \text{ kJ}$$

$$W = Q_2 + Q_1$$

$$= -150 + (250)$$

$$= 250 - 150$$

$$W = 100 \text{ kJ}$$

$$\eta = \frac{W_{\text{net out}}}{Q_{\text{net in}}}$$

$$\eta = \frac{100}{250} \times 100$$

$$\eta = 40\%$$

Problem # 2:

$$P_1 = 300 \text{ kPa}, V_1 = 0.05 \text{ m}^3$$

The gas undergoes an isothermal expansion until its volume doubles. During expansion, the gas absorbs 100 J of heat from surroundings.

(a) $P_2 = ?$

(b) Calculate the work done by the gas during expansion

(c) What's the internal energy (ΔU)?

$$P_1 V_1 = P_2 V_2$$

$$\frac{P_1 V_1}{V_2} = P_2$$

$$P_2 = \frac{(300 \text{ kPa})(0.05)}{0.1}$$

$$\Rightarrow P_2 = 150 \text{ kPa}$$

b) work done by the ^{ideal} gas

$$W = nRT \ln\left(\frac{V_2}{V_1}\right)$$

$$W = P_1 V_1 \ln\left(\frac{V_2}{V_1}\right)$$

$$W = (300\text{K})(0.05) \ln\left(\frac{0.05}{0.1}\right)$$

$$W = 10.4 \text{ kPa}$$

c) For ideal gas under isothermal process
 $\Delta U = 0 \rightarrow$ cause no change in temperature.

Problem #3

$$m = 1100 \text{ kg}$$

$$K.E = 400 \text{ kJ}$$

$$V = ?$$

$$K.E = \frac{1}{2}mv^2$$

$$\sqrt{\frac{400 \times 10^3}{1100}} = v$$

$$v = 26.96 \text{ m/s}$$



2mor

Problem # 3(16)

$$h = ?$$

$$P.E = 400 \text{ kJ}$$

$$m = 1100$$

$$P.E = mgh$$

$$\frac{P.E}{mg} = h$$

$$h =$$

$$\frac{400000}{(1100)(9.8)}$$

$$h = 37.106 \text{ m}$$

Problem # 4

$$m_1 = 10 \text{ kg}$$

$$m_2 = 100 \text{ kg} \rightarrow \text{liquid water in bucket}$$

$$h = 10.2 \text{ m}$$

$$T = \text{same (state 1)}$$

then stone is fall into the water (state 2)

$$\Rightarrow \Delta U, \Delta KE, \Delta PE, Q, W$$

The stone has just rest into the bucket (state 3)

Heat has been transferred to the surrounding

in such a amount that stone & water are at same temperature (state 4)

D

Solution: !

Energy equation

$$DU + DK \cdot E + DP \cdot E_2 = 0 - E \quad (1)$$

- (a) $du = 0$ cause temp. is constant
(b) $d_{1,2} = 0$
(c) $\hat{W}_{\text{work done}} = 0$

$$\begin{aligned} \text{state } 1-2 \quad & \left\{ \begin{aligned} 0 + \Delta K \cdot E + \Delta P \cdot E &= 0 - 0 \\ \Delta K \cdot E + \Delta P \cdot E &= 0 \\ \Delta K \cdot E = -\Delta P \cdot E &= -mg(h_2 - h_1) \\ &= -(10)(9.8)(0 - 10.2) \\ \Delta K \cdot E &= 1 \text{ kJ} \end{aligned} \right. \end{aligned}$$

$$\left. \begin{aligned} \Delta U + \Delta K \cdot E + \Delta P \cdot E &= Q - W \\ \Delta P \cdot E &= 0, \quad Q = 0, \quad W = 0 \\ \Delta U + \Delta K \cdot E &= 0 \\ \Delta U &= -\Delta K \cdot E \\ \Delta U &= -1 \text{ kJ} \end{aligned} \right\} \text{state } 2 \rightarrow 3$$

$$\left. \begin{aligned} \Delta U + \Delta K.E + \Delta P.O.E &= Q - W \\ \Delta K.E &= 0, \Delta P.O.E = 0, W = 0, \Delta U = -1 \text{ kJ} \\ \Delta U &= Q \\ Q &= -1 \text{ kJ} \end{aligned} \right\} \begin{array}{l} \text{state} \\ 3 \rightarrow 4 \end{array}$$

Modes of Heat transfer

Heat:-

Energy transferred across the boundary of the system from one body to another (surrounding) due to temperature difference b/w two system. Heat naturally flows from a system at high temp. to one at a temperature. If heat doesn't flow $\rightarrow$ Thermal Equilibrium

Unit:

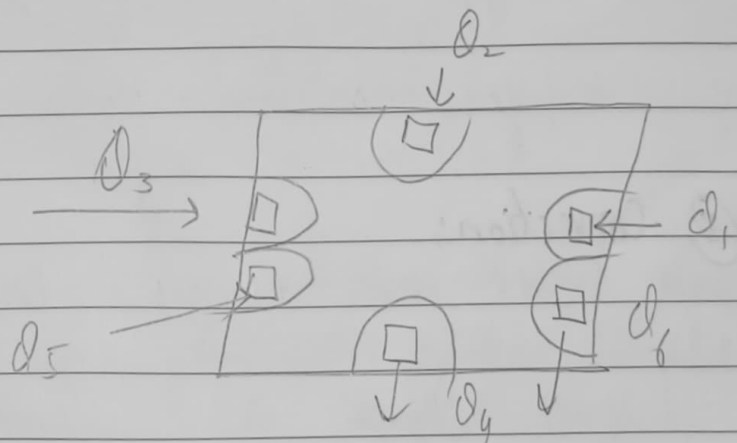
Joule (SI), BTU, Calorie.

Sign Convention:

Heat into the system: +ve
Heat out of the system: -ve

Path function:

$$Q_{\text{total}} = \int \delta Q$$



① Conduction:-

Transfer of heat due to the direct contact b/w particles of matter. It happens because higher energy particles (those at higher temp)



transfer energy to lower energy molecules (those at lower temp.) via molecular collision.

Fourier's law of conduction:

$$\dot{Q} = -KA \frac{dT}{dn}$$

- $\dot{Q}$ = Rate of heat transfer
- K = Thermal conductivity of material
- A = Cross sectional area through heat is transferred
- $\frac{dT}{dn}$ = Temperature gradient

K (for metals): ^{about} 100 W/m.K

K (for insulator): upto 0.01 W/m.K

② Convection:

Heat transfer through a moving fluid (liq or gas). It is a combination of bulk motion of fluid and conduction.

Newton's law of cooling:





$$\dot{Q} = h A \Delta T$$

convective heat transfer coefficient

- ① Natural convection
- ② Forced convection

Radiation:

Heat transfer due to waves and doesn't require any medium.
e.g. Heat from sun that reaches Earth:

- Stefan-Boltzmann law

$$\dot{Q} = E \sigma A T_s^4$$

E = Emissivity of the surface (0 to 1)
 σ = Stefan-Boltzmann constant (5.67×10^{-8})
 A = surface area
 T_s = surface temp

Problem:-

Room temp = $T_{\text{room}} = 20^\circ\text{C}$

Outside temp = $T_{\text{ambient}} = -10^\circ\text{C}$

outer glass surface temp = $T_{\text{outer}} = 12.1^\circ\text{C}$

Glass thickness = $x = 0.005\text{ m}$

$k = 1.4\text{ W/m}\cdot\text{K}$

$A = 0.5\text{ m}^2$

$h = 100\text{ W/m}^2\cdot\text{K}$

→ conduction through glass?
 → convection from glass to ambient air?



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$$Q_{\text{conduction}} = -KA \frac{\Delta T}{\Delta x}$$

$$\Delta T = 20 - 12.1 = 7.9^\circ\text{C}$$

$$\Delta x = 0.005 \text{ m}$$

putting the values

$$Q_{\text{conduction}} = -1106 \text{ W}$$

$$Q_{\text{convection}}: hA\Delta T$$

$$\Delta T = 12.1^\circ\text{C} - (-10^\circ\text{C}) = 22.1^\circ\text{C}$$

Putting values

$$Q_{\text{convection}} = 1105 \text{ W}$$

Extensive VS Intensive properties:

• Extensive properties:

Depends on the amount of matter
(mass, volume, internal energy)

• Intensive properties:

Independent of the mass of the
system (temp, pressure)



WaRQnotes

Internal Energy

$$U = m u$$

(U) → Total internal energy
 (m) → mass
 (u) → specific internal energy

$$u_{fg} = u_g - u_f$$

$$u_{fg} = u_g - u_f$$

Internal energy in the liquid-vapour region.

$$U = U_{liq} + U_{vap}$$

$$u = 0 \text{ (saturated liq.)}$$

$$U = (1 - u) u_f + u u_g$$

$$u = 1 \text{ (saturated vap.)}$$

$$u = u_f + u (u_g - u_f)$$

Enthalpy:-

Enthalpy combines internal ^{energy} and pressure-volume work. It is used while analyzing processes that occurs at constant pressure.

Application involve boilers, turbines and compressors where heat transfer at constant pressure is common (h)

Derivation:

Consider a control mass of gas undergoing equilibrium constant pressure work. The work is only done at boundary of the gas state as it expands or compresses. According to first law of Thermodynamics:

$$U_2 - U_1 = Q_{12} - W_{12}$$

↙
↖

final
initial

We assume a constant-Pressure process, the work is done is given by:

$$W_{12} = P(V_2 - V_1)$$

Energy equation (First law of thermodynamics) becomes:

$$U_2 - U_1 = Q_{12} - P(V_2 - V_1)$$

$$Q_{12} = (U_2 - U_1) + P(V_2 - V_1)$$

Notice that expression for heat transfer involves a combination of internal energy and pressure-volume work. We define a new property called Enthalpy as:

$$H = U + PV$$

For constant pressure process, the heat transfer is:

$$Q_{12} = H_2 - H_1$$

Specific enthalpy: $h = U + PV$

Saturated and superheated state:

h_f : enthalpy of saturated liquid

h_g : enthalpy of saturated vapour

h_{fg} : enthalpy of saturated liquid-vapour mixture

$$h = h_f + n \cdot h_{fg}$$

This is heat during the phase change of water from liquid to vapour (boiling)

Units: kJ/kg

Application:-

→ Boilers and steam turbines

Constant pressure, the heat added

$$Q_{\text{boiler}} = H_{\text{steam}} - H_{\text{water}}$$

Work done by the turbine:

$$W_{\text{turbine}} = H_{\text{inlet}} - H_{\text{outlet}}$$

Heat exchange:

$$Q = \dot{m} \cdot (h_{\text{out}} - h_{\text{in}})$$

$\dot{m}$ is the mass flow rate

Internal Energy:

$$U = h - Pv$$

Enthalpy is a state property, means it only depends on the state of the substance not on the path taken to reach that state

Problem Analysis and Solution Technique

Steps:-

Step 1:

Make a sketch of the system → why?

why?

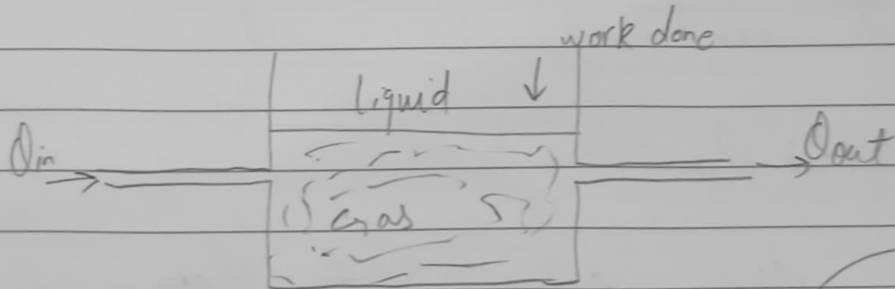
Helps in understanding how different aspect of the system interact.

How?

Indicate the direction of mass flows, heat transfer, work & external forces

Example:

Piston cylinder system, sketch the system, the gas inside and allows show any flows of heat or work done.



Step 2:

Define a control mass or control volume. → specific space through which matter flows

Why?:

Defining the boundary of the system helps in applying the right laws and equations.

How? :

Place a close surface around the system or subsystem you are analyzing. Identify any inflows or outflows of mass, energy and work.

Step 3: Apply General conservation laws

Why? :

There are fundamental to solve thermodynamics problem (conservation of energy, mass first law of thermodynamics)

How:

Energy equation: → ensure all inflows and outflows are mentioned

$$U_2 - U_1 = Q_{12} - W_{12}$$

Step 4: Write down any specific law

Why? :

These allow us to handle any specific details about the substance involved in the system.

(ideal gas or steam etc)

How:

Reference property table and apply specific law based on the nature of the process (constant volume, isothermal, adiabatic etc)

Step 5: Solve using diagram or tables

Why?:

$P-v$, $T-v$ diagram helps visualize the process:

How?:

$P-v$, $T-v$ diagram visualization state determination.

Step 6: Formulate and solve the equation

How?:

Combine all the equations and relationships then solve for unknown.

Example:

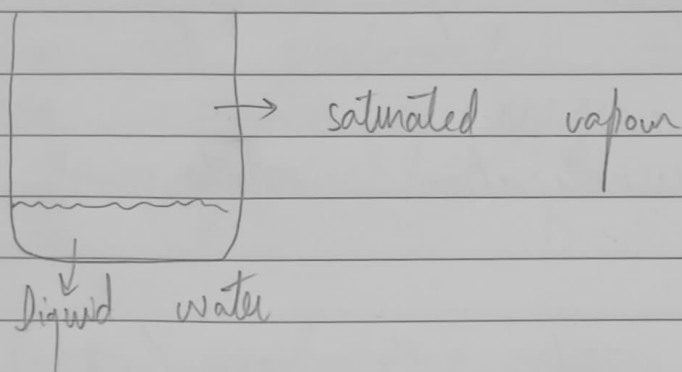
$$U_2 - U_1 = Q_{12} - P(U_2 - U_1)$$

Q:

A vessel with a volume of 5 m^3 contains 0.05 m^3 of saturated liquid water and 4.95 m^3 of saturated vapour at 0.1 MPa . Heat is transferred to the system until it is filled with saturated vapour. Determine the heat transfer for this process.

Solution:

Step 1:



Step 2:

It is a control mass, control volume system constant volume process.

Step 3:

$$U_2 - U_1 = Q_{12} - \cancel{W_{12}} \quad \text{when process is in constant volume process.}$$

step 4:

Initial state

$$v_f = 0.001043 \text{ m}^3/\text{kg} \quad , \quad u_f = 417.36 \text{ kJ/kg}$$

$$v_g = 1.6940 \text{ m}^3/\text{kg} \quad v_g = 2506.1 \text{ kJ/kg}$$

Mass of saturated liquid:

$$m_{\text{lig}} = \frac{V_{\text{lig}}}{v_f} = 47.94 \text{ kg}$$

$$m_{\text{vap}} = \frac{V_{\text{vap}}}{v_g} = 2.92 \text{ kg}$$

$$\text{Total mass} = 50.86 \text{ kg}$$

Conservation of Mass:

"Mass cannot be created or destroyed."

"control volume but no control mass."

For control volume, any change in mass inside the control volume must be accounted for any mass entering or leaving the control surface. The equation of continuity is

$$\frac{dm_{\text{cv}}}{dt} = \sum \dot{m}_{\text{in}} - \sum \dot{m}_{\text{out}}$$



If $\frac{dm_{cv}}{dt} = 0$ then mass remains constant.

Mass flow Rate Equation:

$$\dot{m} = \rho AV$$

$$\text{kg/s} = \text{kg/m}^3 \cdot \text{m}^2 \cdot \frac{\text{m}}{\text{s}}$$

$$\text{kg/s} = \text{kg/s}$$

ρ = fluid density

A = cross sectional Area

V = is the velocity of the fluid perpendicular to the surface.

$$\dot{m} = \frac{VA}{v}$$

$v \rightarrow$ specific volume

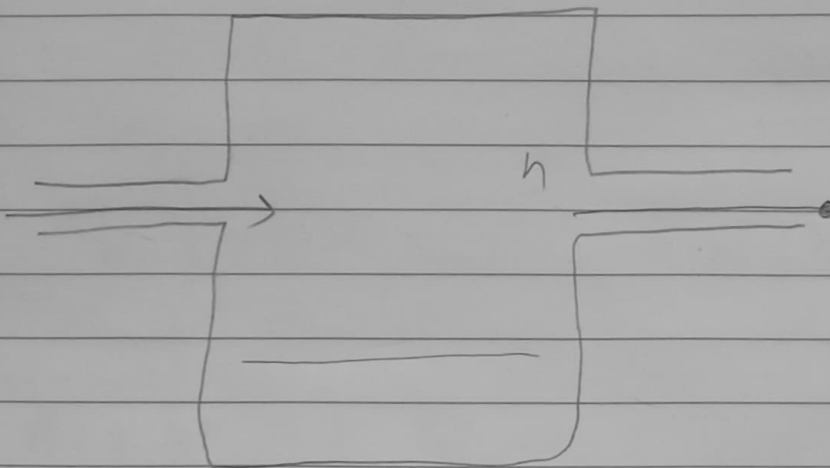
Question:

An ideal gas is flowing $D = 0.2 \text{ m}$, $V = 0.1 \text{ m/s}$
 $T = 25^\circ\text{C}$ $P = 150 \text{ kPa}$ determine $\dot{m}$?

Solution:

$$\left. \begin{aligned} A &= \pi r^2 \\ &= \pi \left(\frac{D}{2}\right)^2 \\ &= \pi \left(\frac{0.2}{2}\right)^2 \end{aligned} \right\} \begin{aligned} A &= 0.0314 \\ PV &= RT \\ V &= \frac{RT}{P} \end{aligned}$$

1- Control volume is stationary boundary of the control volume doesn't move



Unit:-

$$\dot{W} = \frac{J}{s} = \text{Watt}$$

Assumption:-

$$\frac{dm_{cv}}{dt} = 0 \quad \text{and} \quad \frac{dE_{cv}}{dt} = 0$$

Continuity equation:-

$$\sum \dot{m}_i = \sum \dot{m}_e$$

Heat exchanger:-

A steady state heat exchanger is a simple fluid flow through a pipe or system of pipes where heat is transferred to or from the fluid. The fluid may be heated or cooled and may or may not boil, changing from liquid to vapour, or condense, changing from vapor to liquid.

Example #4.3

Analysis

$\sum \dot{m} h_i = \sum \dot{m} h_e$ (steady state)
As there are 2 fluid no,

$$\dot{m}_r (h_i)_r + \dot{m}_w (h_i)_w = \dot{m}_r (h_e)_r + \dot{m}_w (h_e)_w$$

Data:-

$P_1 = 1 \text{ MPa}$ } Refrigerant
 $T_1 = 60^\circ\text{C}$ }

$P_2 = 0.95 \text{ MPa}$ } water
 $T_2 = 35^\circ\text{C}$ }

B.S.2

$$(h_i)_r = 441.89 \text{ kJ/kg}$$

$$(h_e)_r = 249.10 \text{ kJ/kg}$$

$$(h_i)_w = 42.00 \text{ kJ/kg}$$

$$(h_e)_w = 83.95 \text{ kJ/kg}$$

$$\dot{m}_w = \dot{m}_r \frac{(h_i)_r - (h_e)_r}{(h_e)_w - (h_i)_w} = 0.2 \text{ kg/s} \frac{(441.89 - 249.10)}{(83.95 - 42.00)}$$



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Example # 4.4

Chapter # 5

Heat Engines and Refrigerators:-

Heat Engines:-

A device that operates on a thermodynamic cycle, transferring heat from a high temperature reservoir to a low temperature reservoir and producing net positive work. $\Rightarrow Q - W$

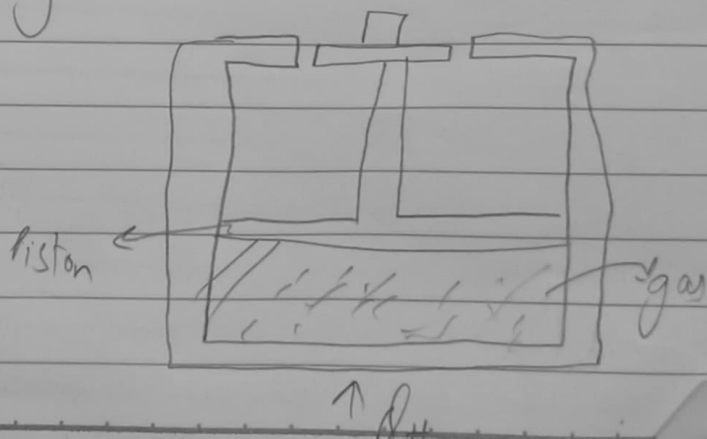
→ Steam, gas or any other substance undergoes a cycle process of heating and cooling.

Steam power plant / engine

→ Heat Q_H is absorbed from a high temperature source

→ Steam flows through the turbine → performing work

→ Steam is condensed, releasing Q_C heat to a low temperature reservoir.



Thermal Efficiency:-

It is a measure of how well the heat engine converts heat engine converts heat input into useful work.

$$\eta_{\text{thermal}} = \frac{W}{Q_H} = 1 - \frac{Q_C}{Q_H} \rightarrow W = Q_H - Q_C$$

Efficiency of real engine:-

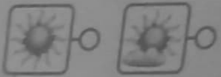
- Large power plant : 35-50 %
- Gasoline engine : 30-35 %
- Diesel Engine : 30-40 %
- Smaller engines : 20 %

Refrigerators and Heat pumps:-

Refrigerator is a device that uses work to transfer heat from a lower temperature to a higher temperature one. It operates on a cycle, similar to a heat pump, while a focus on cooling.

Basic Cycle:-

A typical refrigerator uses a vapour temp cycle.



- Refrigerant R-134a or ammonia absorbs Q_L in the evaporator at low pressure and temperature
- Refrigerant releases Q_H in the condenser

Coefficient of Performance (COP)

$$\beta = \frac{Q_L}{W} = \frac{Q_L}{Q_H - Q_L} = \frac{1}{\frac{Q_H}{Q_L} - 1}$$

Heat Pump:-

When the objective is to heat a space rather than cool it, it is called heat pump.

$$\text{COP} = \beta' = \frac{Q_H}{W} = \frac{Q_H}{Q_H - Q_L}$$

$$\therefore \beta = \frac{Q_L}{Q_H \times \eta}$$

$$\beta = \frac{Q_L \beta'}{Q_H}$$

- Reservoir from which heat is transferred it is called source
- Reservoir to which heat is transferred is called sink

Second Law of thermodynamics:

- Governs energy transfer and conversion in thermodynamics system.
- Introduces concepts of irreversibility and efficiency
- The law has two classical statements
 - ↳ Kelvin Planck statement
 - ↳ Clausius statement

Kelvin Planck Statement:

It is impossible to construct a device that operates in a cycle and produces no effect other than the raising of a weight (work done) and the exchange of heat with a single thermal reservoir.

Key Implications:

- Heat engine can not be 100% efficient
i.e. $\eta \neq 100\%$



$$\eta = \frac{W}{Q_H} \neq 100\%$$

not converted into 100%. Q_H is not possible cause Q_H is

→ Heat must flow from a high temp. to a low temp. reservoir.

Clausius Statement:

"It is impossible to construct a device that operates in a cycle and produces no effect other than the transfer of heat from a cooler body to a warmer body."

→ Refrigerator / Heat pump: Heat cannot flow from cold reservoir to a hot reservoir without work done.

$$COP_{ref} = \frac{Q_L}{W} = \frac{Q_L}{Q_H - Q_L}$$

$$COP_{heatpump} = \frac{Q_H}{W} = \frac{Q_H}{Q_H - Q_L}$$

→ Violation of Clausius statement implies violation of Kelvin-Planck statement

→ The reverse is also true. *Reversible process*

3- Perpetual Motion Machine

First Kind:

creates energy from nothing, violating the first law of thermodynamics

Second Kind:

converts heat from a single reservoir entirely into work, violating the second law.

Third Kind:

operates indefinitely without friction or work output theoretically impossible.

Thermal Reservoir:

- High temp.
- low temp.

→ $T_H - T_C$ (if the difference is higher the efficiency is also higher)

Reversible Process:-

- A reversible process is an idealized process that when reversed leaves both the system and its surroundings completely unchanged.
- In reality, all processes are irreversible.
- Irreversible processes from equilibrium.

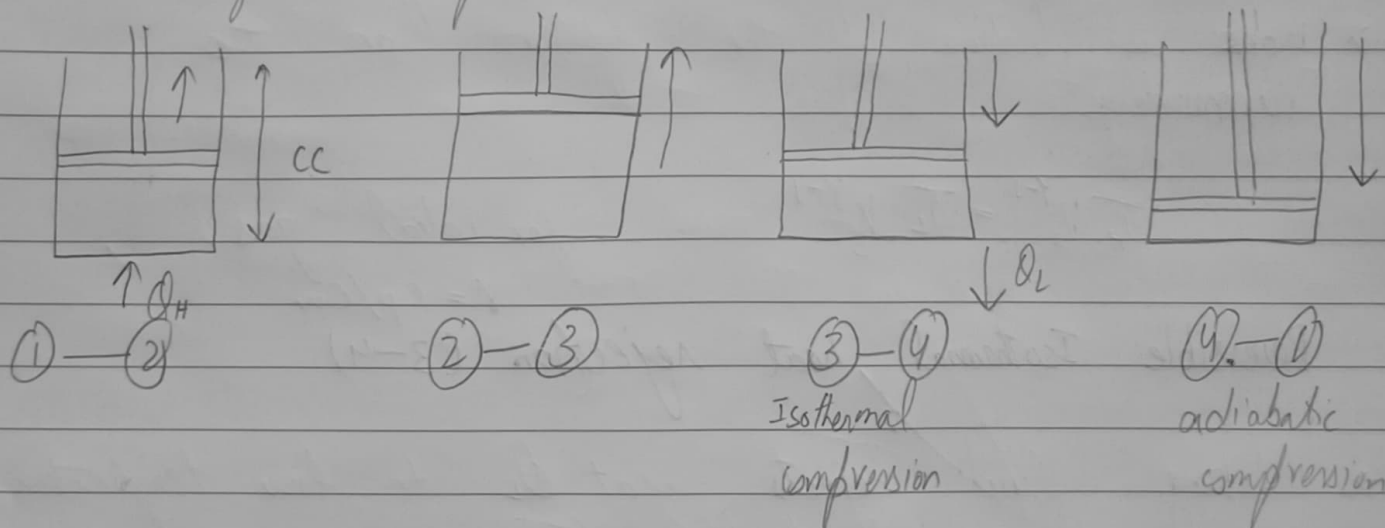
Factor of Irreversibility:

- 1- Friction
 - 2- Unrestrained Expansion
 - 3- Heat transfer through a finite temperature difference
 - 4- Mixing of different substances.
- Internal irreversibility
 - External irreversibility

Carnot cycle

Carnot cycle consist of 4 reversible process

- 1- Reversible Isothermal heat addition at a high temperature
- 2- Reversible adiabatic expansion from the high temperature to the low temperature
- 3- Reversible isothermal heat rejection at a low temperature
- 4- Reversible adiabatic compression back to the high temperature



1- Reversible Isothermal heat addition at a high temperature: (1-2)

- Absorb Q_H from high temperature reservoir
- Heat transfer takes place
- Working fluid expands, doing work on the surrounding.

→ may involve a phase change

$Q_H = T_H (\Delta S) \rightarrow$ change in entropy
→ Q and W are positive

Reversible Adiabatic Expansion (2-3)

- Working fluid expands without exchanging heat with the surrounding.
- Adiabatic process $Q = 0$
- Temperature of the fluid decreases from T_H to T_L
- Work is done by the system on the surroundings

$$T_H V_2^{\gamma-1} = T_L V_3^{\gamma-1}$$

for ideal gas
 $\gamma = C_p/C_v$

Reversible Isothermal heat rejection (3-4)

- Working fluid rejects heat Q_L to low temperature reservoir at constant temperature (T_L)
- Working fluid temperature is infinitesimally higher than T_L
- The process is isothermal
- Heat is removed from working fluid, causing partial condensation or compression
 $Q_L = T_L \Delta S$

Reversible Adiabatic compression (4-1)

- The working fluid is compressed adiabatically, raising temperature back to the condition
- Adiabatic process $Q=0$
- Reversible work done on the system by surrounding
- Effect on system:

system temperature raise from T_L to T_H

$$T_L V_4^{\gamma-1} = T_H V_1^{\gamma-1} \Rightarrow U = +W$$

Efficiency of Carnot cycle:

$$\eta = 1 - \frac{Q_L}{Q_H} = 1 - \frac{T_L}{T_H}$$

key feature:

- Reversible
- Depends only on the temperature
- Entropy

$$\Delta S_H = \frac{Q_H}{T_H}, \quad \Delta S_L = \frac{Q_L}{T_L}$$

Application:

- Real world limitation
- No real engine can operate on the Carnot cycle due to irreversibilities
- Carnot cycle is just a benchmark for real engines

Two properties regarding the efficiency of a Carnot cycle:

- 1- It is impossible to construct an engine that operates between two given reservoirs and is more efficient than a reversible engine operating between the same two reservoirs.

$$\eta_{\text{any}} \leq \eta_{\text{reversible}}$$

→ η_{any} → efficiency of any engine operating between the reservoirs.

- 2- All engines that operate on the Carnot cycle between two given constant-temperature reservoirs have the same efficiency.

$$\eta_{\text{rev}_1} = \eta_{\text{rev}_2}$$

Chapter # 6

Entropy

$$\oint \frac{dQ}{T} \leq 0$$

The inequality applies to all possible thermodynamic cycles - whether reversible or irreversible - and is used to define entropy

Entropy is define as

$$ds = \frac{\delta Q}{T} \Big|_{rev}$$

Entropy change

$$S_2 - S_1 = \int_1^2 \frac{\delta Q}{T} \Big|_{rev}$$

Reversible Heat Engine cycle:-

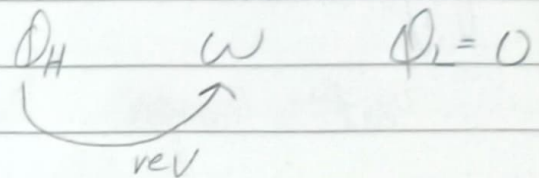
Net heat transfer $\left\{ \delta Q = Q_H - Q_L > 0 \right.$

net entropy change $\oint \frac{\delta Q}{T} = \frac{Q_H}{T_H} - \frac{Q_L}{T_L} = 0$

For all reversible cycles

$$S_2 - S_1 = \oint \frac{\delta Q}{T} = 0$$

Irreversible heat Engine



Work:

$$W_{irr} < W_{rev}$$

Heat transfer:



Net Entropy change

$$\oint \frac{Q}{T} = \frac{Q_H}{T_H} - \frac{Q_L^{irr}}{T_L} < 0$$

For all irreversibly cycle:

$$S_2 - S_1 = \oint \frac{\delta Q}{T} < 0$$

- High entropy mean energy is scattered and things are more disorder / randomness.
- Low Entropy energy is more organized.

When heat flows into the system Entropy increase (+ve) and vice versa.

heat into the system → +ve
heat out of the system → -ve

Refrigerators:-

Heat transfer $\oint \delta Q = -Q_H + Q_L < 0$

Net entropy change $\oint \frac{\delta Q}{T} = -\frac{Q_H}{T} + \frac{Q_L}{T} = 0$

Irreversible Refrigeration cycle:-

$$W_{irr} > W_{rev}$$

$$Q_H^{irr} > Q_H^{rev}$$

Net entropy change

$$\oint \frac{\delta Q}{T} = -\frac{Q_H^{irr}}{T_H} + \frac{Q_L}{T_L} < 0$$

$$\oint \frac{\delta Q}{T} = 0 \quad (\text{reversible cycle})$$

$$\oint \frac{\delta Q}{T} < 0 \quad (\text{irreversible cycle})$$

Absolute zero $\rightarrow -273^{\circ}\text{C} \rightarrow 0$ Kelvin

entropy = 0 (Due to third law of thermodynamics)

Due to the negligible movement of molecules:-

Entropy $\rightarrow 0$

Specific Entropy ($\text{KJ/kg} \cdot \text{K}$)

Saturated region:

$$S = (1-x)S_f + xS_g$$

$$S = S_f + xS_{fg}$$

Absolute zero $\rightarrow -273^{\circ}\text{C} \rightarrow 0$ kelvin

entropy = 0 (Due to third law of thermodynamics)

Due to the negligible movement of molecules:-

Entropy $\rightarrow 0$

Specific Entropy ($\text{kJ/kg}\cdot\text{K}$)

Saturated region:

$$S = (1-x)S_f + xS_g$$

$$S = S_f + xS_{fg}$$



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Ch# 6 - 6.1, 6.2, 6.3, 6.4, 6.6

Date: / /

Problem:-

Water at 100°C , quality 50% in a rigid bon is heated to 110°C . How do the P, V, u, h change P (increase, decrease or stay the same)

Solution:

A fixed mass in a rigid bon

$V \rightarrow \text{constant}$

$P \rightarrow \text{increase}$

$u \rightarrow \text{increase}$

Quality $\rightarrow \text{increase}$ (cause water convert into steam (vapor state))

Entropy $\rightarrow \text{increase}$

Problem:-

a) Determine the entropy for

R-410a, $T = 25^\circ\text{C}$, $V = 0.01 \text{ m}^3/\text{kg}$

b) Determine the missing property among P, T, S and u for R-410a

a) $T = -20^\circ\text{C}$, $V = 0.1377 \text{ m}^3/\text{kg}$

$$u = \frac{(u - u_f)}{u_{fg}}$$

$$V = v_f + x v_{fg}$$

$$x = 0.6377$$



WUOL

$$S = 0.7618$$

b)

$$v_f < v < v_g$$

$u > u_g \rightarrow$ superheated $\rightarrow$ undefined (quality)
 $u \rightarrow$ undefined
 $P \rightarrow 200 \text{ kPa}$

shear

$$S \rightarrow 1.1783 \text{ kJ/kg K}$$

c)

$$P = 200 \text{ kPa}$$

$$S = 1.409 \text{ kJ/kg K}$$

$$T = ? \quad u = ?$$

$$T = 60^\circ\text{C}$$

$u =$ undefined (superheated state)

d)

Determine the missing property among P, v, s and n of CO_2

$$T = 20^\circ\text{C}, \quad S = 1.49 \text{ kJ/kg K}$$

(a)

$u \rightarrow$

$$S > S_g \rightarrow 1.04$$

$$1.49$$

superheated

$$\frac{P - 1400}{2000 - 1400} = \frac{1.49 - 1.5283}{1.4438 - 1.5283}$$

$$P = 1671.9 \text{ kPa}$$

$$\frac{V - 0.03648}{0.02453 - 0.03648} = \frac{1671.9 - 1400}{2000 - 1400}$$

$$V = 0.03106 \text{ m}^3/\text{kg}$$

6.4

Entropy change in reversible process.

(a) Process 1-2: Isothermal expansion

$$S_2 - S_1 = \int_1^2 \frac{\delta Q}{T} = \frac{Q_1}{T_H}$$

On the T-S diagram, area (1-2-b-a-1) represent heat absorb (Q_H)

$$(b) S_2 - S_1 = \int_1^2 \frac{\delta Q}{T} = 0 \quad \because \text{no change in heat.}$$

Process 2-3: Adiabatic Expansion:



$$\Delta S = S_2 - S_1$$

Isentropic process

(c) Process 3-4: Isothermal compression at T_L

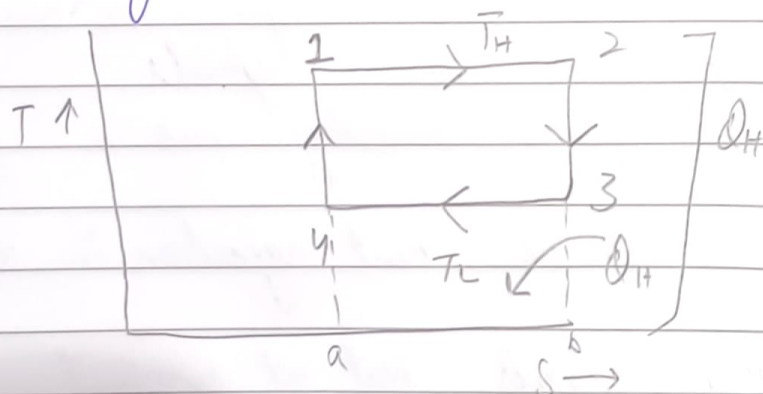
$$S_4 - S_3 = \int_3^4 \frac{\delta Q}{T} = \frac{-Q_3}{T_L} \rightarrow \text{Heat is released causing the entropy to decrease.} \therefore \Delta S \rightarrow -ve$$

(d) Process 4-1: Adiabatic Compression.

$$S_1 - S_4 = \int_4^1 \frac{\delta Q}{T} = 0$$

$$\Delta S = S_1 - S_4 = 0$$

Carnot cycle on Temp. - entropy diagram:



Efficiency & Entropy

$$\eta_{th} = \frac{W_{net}}{Q_H} = \frac{\text{Area of } 1-2-3-4-1}{\text{Area of } 1-2-b-a-1}$$

The Rankine cycle:

1) Process (1-2):

2) Process (2-3): Constant pressure heat addition in the boiler.

→ Heat is transferred to the fluid at a constant pressure in the boiler

→ The process converts the compressor liquid (state 2) into vapours (state 3)

→ If super is involved, the fluid moves to state 3

3) Process (3-4) Reversible Adiabatic expansion in turbine → The superheated vapours expand isentropically in the turbine, providing work.

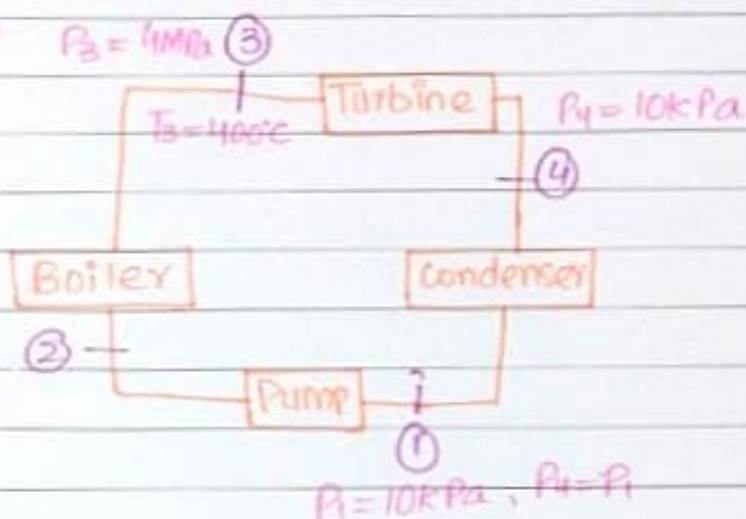
4) Process (4-1) Constant pressure heat rejection in the condenser.

→ The working fluid rejects heat at constant pressure in the condenser

→ The process brings the fluid back to the saturated liquid state at Point 1.

→ Increasing boiler pressure
→ Increases efficiency

Example 9.2



Step 1 Pump work (w_p)

① Inlet state $P_1 = 10 \text{ kPa}$, saturated liquid
 $h_1 = 191.8 \text{ kJ/kg}$, $v = 0.001 \text{ m}^3/\text{kg}$
 $w_p = v(P_2 - P_1) = 4 \text{ kJ/kg}$

$$\Rightarrow w_p = h_2 - h_1 \Rightarrow h_2 = h_1 + w_p = 195.8 \text{ kJ/kg}$$

$$h_3 = 3213.51 \text{ kJ/kg}, s_3 = 6.7690 \text{ kJ/kg}\cdot\text{K}$$

$$q_h = h_3 - h_2 = 3017.8 \text{ kJ/kg}$$

Step 3 $P_4 = 10 \text{ kPa}$
 $s_4 = s_3 = 6.7690 \text{ kJ/kg}\cdot\text{K}$
 $s_4 = s_f + x s_{fg}$
 $x_4 = 0.8159$
 $h_4 = 2144.1 \text{ kJ/kg}$

→ For heat additions

9.3 Effect of Pressure and Temp on Rankine Cycle

① Reverse adiabatic compression in the pump

($\Delta S = 0$, $S \rightarrow \text{const}$, $Q = 0$)

② Isobaric (const Pressure) heat addition ($P \rightarrow \text{const}$)

③ Reversible adiabatic expansion in the turbine

($\Delta S = 0$, $Q = 0$, $S \rightarrow \text{const}$)

④ Isobaric (Const pressure) heat rejection ($P \rightarrow \text{const}$)

Efficiency of the Rankine cycle is influenced by:

- Condenser Pressure (P_4)
- Superheat temp in the boiler (T_3)
- Maximum boiler pressure

Effect of Pressure and Temperature

→ Lower condenser pressure

↳ Increases efficiency

↳ Increases turbine exhaust moisture content

→ Superheating the steam (boiler)

↳ Increases the efficiency

↳ Improves the turbine exhaust moisture content

9.2 The Rankine Cycle

1- Process (1-2):

2- Process (2-3): Constant pressure heat addition in the boiler

→ Heat is transferred to the fluid at a constant pressure in the boiler.

→ The process converts the compressor liquid (state 2) into vapours (state 3)

→ If superheating is involved, the fluid moves to state 3

3- Process (3-4): Reversible Adiabatic expansion in turbine

→ The superheated vapour expands isentropically in the turbine, producing work

4- Process (4-1): Constant-Pressure heat rejection in the condenser

→ The working fluid rejects heat at constant pressure in the condenser.

→ The process brings the fluid back to the saturated liquid state at point 1.

Efficiency of Rankine Cycle:

$$\eta_{th} = \frac{W_{net}}{Q_H} = \frac{\text{Area}(1-2-2'-3-4-1)}{\text{Area}(a-2-2'-3-b-a)}$$

Efficiency depends on